Roll No. | | | | | | | | | | |

Total No. of Pages : 02

Total No. of Questions : 13

B.Pharmacy (Sem.-7)

INSTRUMENTAL METHODS OF ANALYSIS-THEORY

Subject Code : BP-701T

M.Code : 78387

Date of Examination : 11-06-2024

Time : 3 Hrs.

Max. Marks : 75

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains THREE questions carrying TEN marks each and student has to attempt any TWO questions.
3. SECTION-C contains NINE questions carrying FIVE marks each and student has to attempt any SEVEN questions.

SECTION-A

1. **Write briefly :**

(a) Define auxochrom. Give two examples.

(b) *Electrophoretic mobility of an analyte is slow along the wall of capillary than in the center of the capillary.*Why?

(c) What is Rate theory of chromatographic separation?

(d) What is tyndal effect? Give its application in spectroscopic analysis.

(e) Define an overtone band. Give an example.

(f) What are spectral and natural band widths?

(g) How fluorescence differs from phosphorescence?

(h) Name any four solute-property specific detectors used in HPLC.

(i) What are the advantages of HPLC over GC?

(j) What are the different methods of excitation in atomic spectroscopy?

SECTION-B

2. Give detailed description of a UV-VIS spectrophotometer with the help of a neatly drawn labelled flow diagram. Explain in detail various detectors used in it. (4+6)

3. What is the principle of IR spectroscopy? Explain different types of vibrations detected in this technique. How does the vibrational frequency of any bond is affected by different factors? Support your answer with relevant examples. (2+2+6)

4. Write a descriptive note on ion-exchange chromatography with emphasis on its principle, ion- exchange mechanism, different ion-exchange resins and their applications. (2+2+6)

SECTION-C

5. Write a descriptive note on factors affecting fluorescence.

6. Write a brief account on light sources used in an atomic absorption spectrophotometer.

7. What is the importance of atomizer in flame emission spectroscopy? Recall various types of atomizers.

8. What is the principle of electrophoresis? Describe the working, limitations and advantages of capillary electrophoresis.

9. What is the need of derivatization in GC? Discuss various derivatization methods with examples.

10. Write a note on theory, instrumentation and applications of affinity chromatography.

11. How does normal-phase chromatography differ from reverse-phase chromatography? Recall various reverse-phase stationary phases with their specific applications.

12. Derive Beer-Lambert's Law. Discuss different types of deviations associated with its disobeyance.

13. Write a descriptive note on Van-Deemter equation for chromatographic separation.

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NOTE : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.